

Electoral Judicialization and Municipal Competition in Brazil

Evidence from a Bartik Shift-Share Instrument

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Does Judicial Intervention Shape Electoral Competition?

Research question:

Does pre-election judicial intervention causally affect the competitive dynamics of mayoral races?

The mechanism:

- Candidates are sued in electoral courts *before* election day
- Lawsuits impose legal costs, reputational exposure, and campaign disruption
- This could shift vote shares, alter entry decisions, and change voter behavior

The challenge:

Close races *attract* more lawsuits — endogeneity goes in both directions

This Paper

- 5,560 Brazilian municipalities
- 2020 and 2024 mayoral elections
- TSE administrative data: judicial records + electoral results
- Identification: **Bartik shift-share IV**

Key Findings

Two instrument variants ($F \approx 10\text{--}20$). Directionally consistent winner consolidation; blank vote rate rises significantly under the broader specification. Voter participation: precise nulls.

Brazil's Electoral Justice System:

- *Tribunal Superior Eleitoral* (TSE) + 27 *Tribunais Regionais Eleitorais* (TREs) + 2,800+ first-instance zones
- Adjudicates eligibility, campaign finance, conduct *before* election day
- Tens of thousands of cases per cycle — among the world's most active electoral courts
- Rich variation: topic mix differs by municipality, state, and year

Municipal Setting

- 5,570 simultaneous contests = large sample
- Homogeneous office: directly elected mayor
- Uniform electoral rules (Constitution Art. 29)
- Comparable competitive dynamics across municipalities

Two Election Cycles

2020 (Nov 15) → **shares**

2024 (Oct 6) → **treatment + outcomes**

- 1 Motivation
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- 7 Results: Electoral Competition
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Three-tier judicial structure:

- 1 **Zona Eleitoral (ZE)** — first instance; 2,800+ nationwide
- 2 **Tribunal Regional Eleitoral (TRE)** — 27 state appellate courts
- 3 **Tribunal Superior Eleitoral (TSE)** — national apex

Pre-election jurisdiction:

- Candidate registration challenges (*impugnação*)
- Abuse of economic/political power
- Campaign finance violations
- Prohibited propaganda; vote-buying
- Candidate conduct (*conduta vedada*)

Election Timeline

- 1 Candidate registration (August)
- 2 Pre-election lawsuits (Aug–Oct)
- 3 **Election Day** (Oct/Nov)
- 4 Post-election proceedings
- 5 Cassation if ruling goes against winner

2020: Election day = November 15

2024: Election day = October 6

We use only cases filed *before* these dates.

Effects on targeted candidates:

- **Candidacy at risk:** court may rule ineligible before election day
- **Reputational cost:** public filing, media coverage, stigma
- **Resource diversion:** legal defense competes with campaigning
- **Quality signal to voters:** a respondent may be perceived as ethically challenged — or as a victim of political harassment

Who files?

Opposing candidates, parties, *Ministério Público Eleitoral*, any citizen

Treatment Variable

Adversarial competition-relevant cases:

- Mayoral candidate is the *respondent*
- Filed before election day
- Substantive challenge (not mandatory administrative filing)

Measurement:

$\Delta \log(1 + \text{lawsuits}_m)$
from 2020 to 2024

Administrative vs. Adversarial Electoral Litigation

Not all electoral case subjects involve genuine legal challenges to candidates:

Candidacy Registration Confirmation (RRC):

Automatically generated for *every* registered candidacy. No plaintiff, no adversarial content, no legal risk. Purely administrative processing.

Coalition/Party Slate Registration (DRAP): Filed by each party or coalition before every election; mandatory under electoral law. No opposing party, no discretion. Universal coverage (>97% of municipalities in 2020).

Adversarial challenges: Filed by opponents, the Electoral Public Prosecutor (*MPE*), or citizens against a specific candidate's

Case Volume (2020)

Type	Volume	Incl.?
Candidacy registration (RRC)	~994k	×
Coalition registration (DRAP)	~137k	×
Eligibility challenges		✓
Abuse of power		✓
Campaign conduct		✓
Propaganda violations		✓
Information violations		✓

Instrument scope

We test two variants: one that excludes only the mandatory filings, and one (preferred) that retains only adversarial subjects.

- **Restriction:** we keep only cases filed *strictly before* election day:

filing date < election date

- **Rationale:** only pre-election cases can plausibly affect the upcoming competitive dynamics through legal costs, reputational signaling, or candidacy threats
- Post-election *cassação* proceedings determine whether the *realized winner* is removed — a distinct phenomenon that could reverse causality
- This restriction is applied identically to both the 2020 baseline (for shares) and the 2024 measurement (for treatment and shifts)

OLS is likely biased:

1 Competition → lawsuits

Close races attract more suits (parties fight harder); positive confound with competitiveness

2 Vulnerability → lawsuits

Incumbent weakness invites challengers to file; negative confound with incumbency survival

3 Unobservables

Local political culture, partisan intensity, lawyer density — correlated with both

Direction of bias: **ambiguous**

Identification Need

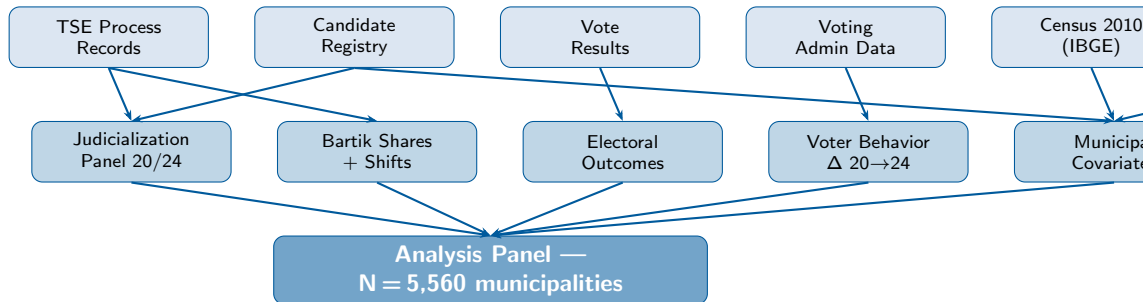
We need exogenous variation in the *level* of judicialization that is independent of the municipality's own competitive dynamics in 2024.

Our Strategy

Bartik shift-share IV:

Predicted 2024 lawsuits based on the municipality's 2020 topic mix interacted with how each topic grew statewide — a source of variation the municipality itself does not drive.

Data Pipeline Overview



Sources (2020 and 2024, publicly available from TSE):

- **Case metadata records:** case class, subject, filing date, originating court zone
- **Case party records:** link each case to its parties (plaintiff and respondent); joined to candidates via voter registration ID
- **Candidate registry:** party affiliation, office sought, municipality, demographics

Linkage chain:

case → respondent party → candidate → office

Keep only mayoral candidates (*prefeito*) as respondents

Key Information Used

Field	Content
Case class	Procedural vehicle
Subject (topic)	Legal subject matter
Filing date	For timing restriction
Party role	Respondent identification
Voter ID	Candidate join key
State / municipality	Geographic identifier

All data publicly released by the TSE under the Brazilian Access to Information Law.

Source: TSE official vote return records, by electoral zone and candidate **Processing steps:**

- 1 Keep first-round results — mayoral races
- 2 Aggregate electoral zones → municipality totals
- 3 Rank candidates by vote share within municipality
- 4 Merge 2020 and 2024 files; compute Δ outcomes

Second round: Only ~222 municipalities (>200k registered voters) eligible. We focus on **round-1 outcomes**, decisive for 97.6% of municipalities.

Outcome Variables

Competition (2024 – 2020):

- Δ runner-up vote share
- Δ winner–runner-up margin
- Δ winner vote share
- Δ probability winner exceeds 50%

Incumbency (level, 2024):

- Incumbent ran in 2024
- Incumbent won in 2024
- Party switch 2020→2024

Extended:

- Δ winner ideology
- Pedersen electoral volatility
- Δ female vote share / winner gender

Source: TSE voter participation aggregates, per electoral zone

Processing:

- Aggregate zones → municipality level
- Compute rates for 2020 and 2024
- Difference: $\Delta_{2024-2020}$

Variables and Baseline Means

Variable	2020	2024
Turnout rate	82.6%	83.5%
Null vote share	4.1%	3.3%
Blank vote share	1.8%	1.9%

All rates denominated on registered voters

Source	Variables	Notes
IBGE Census 2010	Log population; urban share; log income per capita	Tract-level data aggregated to municipality; microregion code extracted
TSE Candidate Registry 2012–2024	Prior candidacies; prior wins per candidate	Linked via voter registration ID; aggregated to municipality $\times$ year
TSE Vote Returns 2016	Margin; effective number of parties; winner party	Clean pre-treatment period for baseline controls
TSE Voter Participation 2020	Turnout; null share; blank share (baseline)	Voter behavior in the base year
TSE Candidate Registry 2020/2024	Gender; education; ethnicity composition	Candidate-pool characteristics by municipality
IBGE Microregiões	558 microregion cells	Used in microregion fixed-effect specification

- **Universe:** 5,570 Brazilian municipalities with mayoral races in both 2020 and 2024
- **Share restriction:** must have ≥ 1 adversarial competition-relevant case in 2020 to receive non-trivial Bartik shares. Zero-case municipalities receive zero instrument value (included but contribute no identification)
- **Analysis sample after listwise deletion of controls: 5,560 municipalities**
- **Population subsamples** based on 2024 registered voters:
 - $\leq 200,000$ registered voters: 5,338 municipalities — single-round regime (Art. 29-II CF/88)
 - $> 200,000$ registered voters: 222 municipalities — second-round eligible
- **Single-zone subsample:** 5,371 municipalities (96.6%) — exactly one electoral zone; cleanest clustering

Summary Statistics

Variable	Mean	SD	Min	Max
<i>Treatment (adversarial-only specification, excl. mandatory filings)</i>				
$\Delta \log(1 + \text{adversarial lawsuits})$	-0.062	0.649	-6.2	5.1
Bartik IV — adversarial-only (Z_m^B)	-0.137	0.161	-1.3	0.4
<i>Outcome variables</i>				
Δ Runner-up vote share	-0.012	0.159	-0.500	0.499
Δ Margin (winner—runner-up)	0.060	0.325	-1.000	1.000
Δ Winner vote share	0.048	0.191	-1.000	0.718
Δ Turnout rate	0.010	0.031	-0.158	0.149
<i>Pre-determined controls</i>				
Log population (2010)	9.41	1.15	6.69	16.23
Urban share (2010)	0.080	0.144	0	1
Log income per capita (2010)	0.42	0.51	0.00	2.39
Margin 2016 (pp)	18.5	21.2	0.0	100.0
<i>N = 5,560 municipalities (baseline analysis sample)</i>				

In 2020, each municipality has a topic mix:

Some had mostly candidate eligibility challenges; others had more campaign conduct or propaganda violations.

By 2024, adversarial litigation waves diffused unevenly across topics.

Waves spread through TRE jurisprudence, party legal strategies, and MPE enforcement patterns — systemic forces that are state-wide, not municipality-specific.

The instrument:

Z_m^B = municipality's *exposure* to these waves, predicted from its 2020 topic mix $\times$ state-level growth in each topic.

How much of the statewide adversarial wave was this municipality structurally positioned to absorb?

Key Design Logic

- 1 Shares:** from 2020 — pre-determined, before 2024 politics materialize
- 2 Shifts:** state-level waves driven by institutional and strategic diffusion, not by any single municipality's 2024 dynamics
- 3 Product:** predicted exposure to adversarial litigation waves, exogenous to local competitive outcomes

$$Z_m^B = \sum_{k \notin \text{admin.}} s_{m,k,2020} \cdot g_{\text{UF}(m), k, -m}$$

$s_{m,k,2020}$	Municipality m 's share of topic k among competition-relevant lawsuits in 2020; re-normalized after exclusion of administrative filings so that $\sum_k s_{m,k,2020} = 1$
$g_{\text{UF},k,-m}$	Leave-one-out (LOO) log growth of topic k in state $\text{UF}(m)$ from 2020 to 2024: $g = \log\left(\frac{T_{\text{UF},k,2024} - C_{m,k,2024}}{T_{\text{UF},k,2020} - C_{m,k,2020}}\right)$
k	Topic = legal subject matter; competition-relevant subjects only
LOO	Municipality m excluded from its own state shift to prevent mechanical correlation

Topic Selection: Two Instrument Variants

We construct two versions of the Bartik instrument, differing only in which case subjects contribute to the shares:

Broad — all competition topics: Includes all 31 competition-classified subjects from the TSE taxonomy, *including* the mandatory administrative filings (RRC and DRAP). 31 topics, $F \approx 20$.

Narrow — adversarial only (preferred): Excludes mandatory candidacy registration (RRC) and mandatory coalition slate registration (DRAP). Retains 29 genuinely adversarial subjects. $F \approx 10$.

DRAP exclusion is motivated by the same institutional logic

Instrument Comparison

Variant	Topics	Nonzero	F
All topics	31	4,045	19.6
Adversarial	29	4,042	10.4

Mean Bartik: -0.12 / -0.14

SD: 0.08 / 0.16

Preferred specification

Adversarial-only is theoretically preferred. The broader variant is reported as a sensitivity check. Direction is consistent across both.

Construction:

- 1 Count unique adversarial cases per (municipality, topic) in 2020
- 2 $s_{m,k,2020} = C_{m,k,2020} / \sum_k C_{m,k,2020}$
- 3 Shares sum to 1.0 per municipality by construction
- 4 Municipalities with zero adversarial 2020 cases: assigned zero instrument value

Granularity: TSE legal subject codes give ~ 29 adversarial topic cells in the preferred specification

Why 2020 Shares?

Shares are **pre-determined**: based on the 2020 case distribution, two electoral cycles before the 2024 election dynamics unfold.

Exposure $s_{m,k,2020}$ cannot be chosen or manipulated by actors anticipating 2024 competition — a core validity argument.

Construction:

- 1 Compute state $\times$ topic totals: $T_{UF,k,t}$ for $t \in \{2020, 2024\}$
- 2 Remove m 's own contribution (LOO): subtract $C_{m,k,t}$
- 3 Compute log growth; set undefined if LOO denominator ≤ 0

What the shifts capture:

State-level growth in topic k reflects waves of adversarial litigation spreading through the institutional and strategic networks of Brazilian electoral politics — TRE chamber interpretations, coordinated party legal strategies, and MPE enforcement priorities — that are common to a state but not driven by any individual municipality.

Design Choices Defended

State level Captures the relevant institutional unit (TRE jurisdiction); closer comparison than national but broader than municipality

LOO Prevents m from mechanically driving its own shift

Admin. out Adversarial litigation only; mandatory filing noise excluded

Log Proportional changes; symmetric treatment of growth and decline

Formal condition:

$$E[Z_m^B \cdot u_m \mid \mathbf{X}_m, \delta_{UF}] = 0$$

Core argument — network diffusion framing:

The instrument does *not* claim that litigation growth is purely random. Rather, Z_m^B measures a municipality's structural **exposure** to adversarial litigation waves that originate in and diffuse through the *institutional and strategic networks* of Brazilian electoral politics — TRE chamber jurisprudence, coordinated party legal strategies, MPE enforcement priorities — operating at the state level. These systemic forces are orthogonal to the specific competitive environment of any individual municipality, conditional on state FE and pre-determined controls.

Potential Threats

- 1 **Large-municipality dominance:** municipality m drives state shift
→ mitigated by LOO; most municipalities are small
- 2 **2020 shares reflect 2020 competition:** shares correlated with unobserved persistence
→ conditional on state FE + 2020 competitive controls
- 3 **Network spillovers:** litigation waves may correlate with state-level political events affecting all municipalities
→ state FE absorbs common state shocks

We verify:

- 1 **Relevance:** first-stage F-statistic; target ≥ 10
Adversarial-only: $F \approx 10.4$
Broad variant: $F \approx 19.6$
- 2 **Balance:** regress Z_m^B on each pre-determined covariate (one at a time, state FE); expect $p > 0.10$ for most
- 3 **Distribution:** no implausible outliers in Z_m^B after residualization on state FE
- 4 **Stability:** first-stage coefficient stable across control sets, samples, and FE granularity
- 5 **Placebo:** mandatory administrative subjects (DRAP, RRC) generate $F \approx 0$ when used alone — confirming the competition-subject selection is doing real work

Benchmark Design

Ash, Morelli & Vannoni (2025, JPE):
Bartik instrument for state legislative detail.

Implementation: `R fixest::feols()`
— SE clustered by principal electoral zone.

Stock–Yogo threshold

Critical value for 10% size distortion with one excluded instrument: $F \approx 10$.

Adversarial-only ($F = 10.4$): borderline adequate.

Broad ($F = 19.6$): comfortably above threshold.

Both reported for transparency.

Which litigation topics actually drive the instrument? (Goldsmith-Pinkham, Sorkin & Swift 2020, AER)

The 2SLS estimand decomposes as $\hat{\tau}_{IV} = \sum_k \alpha_k \hat{\tau}_k$, where $\alpha_k = (\tilde{z}'_k \tilde{d}) / (\tilde{Z}' \tilde{d})$ is topic k 's share of the first-stage numerator and $\hat{\tau}_k$ is the just-identified IV using topic k alone.

Rank	Topic	α (%)	Cum.	$\hat{\tau}_k^{\text{maj.}}$	$\hat{\tau}_k^{\text{fem.}}$
1	Impugn. Registro Candidatura	+42.9	0.43	+0.077	-0.083
2	Recurso Registro Candidatura (RRCI)	+22.6	0.65	+0.195	-0.072
3	Preenchimento Vaga Remanescente	+16.0	0.81	+0.121	-0.053
4	Propaganda Internet	+8.3	0.89	+0.206	-0.090

Top 4 (of 29 adversarial topics) +89.8

$\hat{\tau}_k^{\text{maj.}}$: just-identified IV for Δ winner majority; $\hat{\tau}_k^{\text{fem.}}$: for Δ female vote share.

HHI = 0.27; 24 of 29 topics have positive weight. Residualized on state FE + 7 baseline controls.

What drives the instrument

Top 3 topics are registration challenges (Impugnação + RRCI + Vaga Remanescente) — procedures that disqualify or replace candidates before the election. Together $\approx 82\%$ of first-stage weight.

Sign consistency

All four top topics yield *positive* just-identified estimates for winner majority and *negative* for female vote share — consistent sign across topics supports the aggregate result.

First stage (OLS):

$$\underbrace{\Delta \log(1 + \ell_m)}_{\text{endogenous}} = \alpha + \beta Z_m^B + \mathbf{X}_m' \gamma + \delta_{\text{UF}(m)} + \varepsilon_m$$

Second stage (2SLS):

$$\Delta Y_m = \alpha + \tau \Delta \widehat{\log(1 + \ell_m)} + \mathbf{X}_m' \gamma + \delta_{\text{UF}(m)} + u_m$$

- ℓ_m = adversarial competition-relevant lawsuits (passive party, pre-election)
- Δ = difference 2024 – 2020 for flow outcomes; level in 2024 for cross-section outcomes
- $\mathbf{X}_m$ = pre-determined controls (7 baseline or 14 extended)
- $\delta_{\text{UF}(m)}$ = state or microregion fixed effects
- SE clustered by principal electoral zone

Six Specifications

Specification	Controls	Fixed Effects	Sample	<i>N</i>
baseline_state_fe	7 baseline	State (27)	All	5,560
baseline_sz	7 baseline	State (27)	Single-zone	5,371
robust_full	14 extended	State (27)	All	5,560
sub_1e200k	7 baseline	State (27)	$\leq 200k$ voters	5,338
sub_gt200k	7 baseline	State (27)	$> 200k$ voters	222

7 baseline controls: Census 2010 (log pop, urban share, log income pc) + margin 2016 +

2020 electoral baseline (log valid votes, margin top1–top2, log candidates)

14 extended controls: baseline + ENP 2016 + candidate composition 2020 (gender, ethnicity, education) + party/coalition count + voter behavior baseline + candidate experience

Cluster level: principal electoral zone
(lowest-numbered zone serving the municipality)

- Brazil has 2,800+ *zonas eleitorais*
- 97.6% of municipalities span exactly one zone
- The zone is the unit at which legal filings and candidate lists are administratively organized
- Plausible locus of spatial correlation in political networks and legal strategies

Result: 2,187 clusters in the full sample — far more conservative than 27-state clustering

Cluster Counts by Sample

Sample	Clusters
All municipalities	2,187
Single-zone only	2,049
$\leq 200k$ voters	2,017
$> 200k$ voters	222

Single-zone municipalities: cluster =
municipality = zone (exact correspondence).
Cleanest inference design.

First-Stage Results — Two Instrument Variants

Panel A: Broad instrument (all competition topics, incl. mandatory filings)

Specification	Coef.	SE	<i>t</i>	F-stat	<i>N</i>
Baseline (state FE)	1.792	0.405	4.42	19.6	5,560
Baseline, single-zone	1.801	0.422	4.27	18.2	5,371
Full controls	1.799	0.406	4.44	19.7	5,560
≤200k voters	1.869	0.426	4.39	19.3	5,338
>200k voters	0.799	0.779	1.03	1.1	222

Panel B: Narrow instrument

(adversarial only, excl. RRC + DRAP) — preferred

Specification	Coef.	SE	<i>t</i>	F-stat	<i>N</i>
Baseline (state FE)	0.481	0.149	3.23	10.4	5,560
Baseline, single-zone	0.497	0.155	3.21	10.3	5,371
Full controls	0.491	0.151	3.26	10.7	5,560
≤200k voters	0.493	0.155	3.19	10.2	5,338
>200k voters	0.485	0.566	0.86	0.7	222

SE clustered by principal electoral

zone. $F\text{-stat} = (t\text{-stat on excluded instrument})^2$.

Coefficient interpretation:

A 1-unit increase in Bartik exposure predicts a 0.48-unit increase in $\Delta \log(1 + \text{adversarial lawsuits})$ under the preferred (narrow) specification. The broader variant implies a 1.79-unit increase.

The coefficient difference reflects measurement: the narrow endogenous variable is more dispersed ($SD = 0.65$) than the broad one ($SD = 0.43$), so the same shift in predicted exposure maps to a different log-scale coefficient.

Instrument strength:

Adversarial-only ($F = 10.4$) is borderline at the Stock–Yogo 10% size distortion threshold. Broad variant ($F = 19.6$) is robust. Both stable across control sets.

Stability Check

First-stage F-statistic across specs:

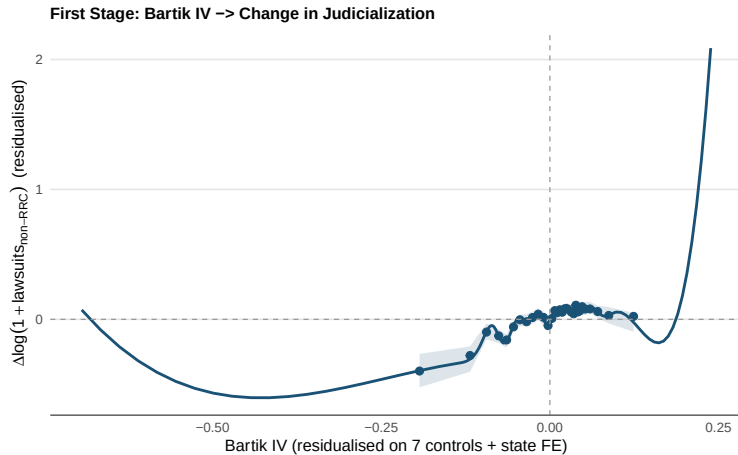
- Broad: $F \in [18.2, 19.7]$ (excl. large cities)
- Narrow: $F \in [10.2, 10.7]$ (excl. large cities)

Consistent with a clean instrument. Large-city subsample ($>200k$) is underpowered in both variants ($F < 2$) and should not be interpreted.

Why narrow F is lower

Excluding DRAP removes a topic with near-universal coverage and a moderate negative shock — it contributed substantial

First Stage — Binscatter



30 equal-count bins. Cubic polynomial fit + 95% CI. Slope ~ 1.44 (OLS residuals). $N = 5,560$.

Electoral Competition — Both Variants (Baseline Specification)

Outcome: $\Delta(\dots)$, 2024 – 2020. $N = 5,560$, clusters = 2,187, state FE + 7 controls.

Outcome	Broad (all topics, $F = 19.6$)			Narrow / preferred ($F = 10.4$)		
	Coef.	SE	p	Coef.	SE	p
Δ Runner-up vote share	−0.031	0.022	0.157	−0.019	0.026	0.456
Δ Margin (W−RU)	+0.062	0.043	0.151	+0.057	0.051	0.268
Δ Winner vote share	+0.031	0.025	0.213	+0.038	0.030	0.207
Δ Winner majority	+0.140	0.090	0.122	+0.113	0.100	0.259
Δ Log n candidates	−0.066	0.039	0.092	−0.052	0.047	0.265

Direction consistent across variants

All five outcomes point in the same direction under both specifications. Competition moves toward winner concentration; candidate count falls. Neither variant reaches $p < 0.05$.

Precision gap

Lower F-stat in the narrow spec translates to wider confidence intervals. The direction of the adversarial-only result is the same as the broad, lending credibility to the pattern.

Consistent pattern across both variants:

Judicialization $\uparrow$
 $\Rightarrow$ **Winner majority** $\uparrow$
(+14 pp, $p = 0.12$)
 $\Rightarrow$ Winner vote share $\uparrow$
(+3.1 pp, $p = 0.21$)
 $\Rightarrow$ Runner-up vote share
 $\downarrow$ (-3.1 pp, $p = 0.16$)
 $\Rightarrow$ Electoral margin $\uparrow$
(+6.2 pp, $p = 0.15$)
 $\Rightarrow$ Number of candidates $\downarrow$ ($p = 0.09$)

Winner consolidation hypothesis

Broad specification shown. Direction identical in adversarial-only.

Mechanism Hypothesis

Adversarial lawsuits may fall disproportionately on front-runner challengers (more to challenge) or deter fringe entry; legal uncertainty resolves in favor of the most established candidate; winner consolidates vote.

Or: lawsuits serve as a quality signal that reduces voter uncertainty, causing vote concentration around the cleaner candidate.

Statistical Status

Directionally consistent across both variants and all control specifications.

No estimate reaches $p < 0.05$ after

Voter Behavior — Both Variants (Baseline Specification)

$N = 5,560$, state FE + 7 controls. All rates denominated on registered voters.

Outcome	Broad ($F = 19.6$)			Narrow / preferred ($F = 10.4$)		
	Coef.	SE	p	Coef.	SE	p
Δ Turnout rate	-0.006	0.005	0.185	-0.008	0.007	0.281
Δ Null rate	+0.007	0.006	0.220	-0.003	0.005	0.509
Δ Blank rate	+0.013	0.005	0.018	+0.004	0.004	0.340
Δ Valid vote rate	-0.026	0.013	0.038	-0.008	0.012	0.485

- **Blank rate (broad):** +1.3 pp, $p = 0.018$ — significant under the broader specification. Municipalities more exposed to diverse adversarial litigation cast more blank votes.
- **Valid vote rate (broad):** -2.6 pp, $p = 0.038$ — consistent with blank rate pattern (accounting decomposition).
- **Turnout and null rate:** precise nulls across both variants — no mobilization or protest-voting channel.
- **Narrow spec:** blank rate attenuates to +0.4 pp, $p = 0.34$ — suggests the blank-rate effect is driven partly by mandatory administrative filings in the topic mix, or reflects a genuine sensitivity to the breadth of judicial activity.

Turnout ($p \approx 0.18$ – 0.28):

Electoral judicialization does not change whether citizens participate. No mobilization or demobilization effect in either variant.

Null (spoiled) rate ($p > 0.20$):

Voters do not express frustration by spoiling their ballot. No protest-vote channel through null votes.

Blank rate ($p = 0.018$ in broad variant):

Significant signal under the broader specification: more diverse judicial activity may generate voter ambivalence, leading to expressive blank voting. Not robust in the adversarial-only spec.

Valid vote rate:

Mirror image of blank rate; no independent information.

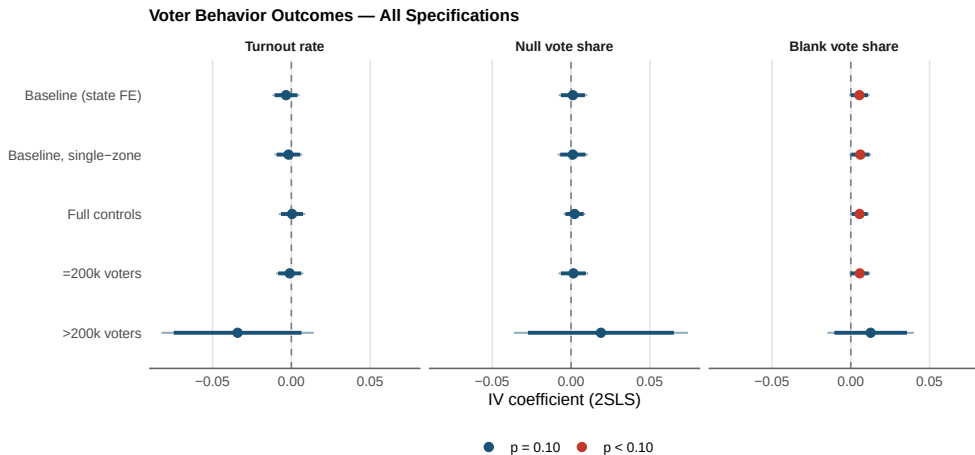
Summary

Outcome	Broad p	Narrow p
Turnout	0.185	0.281
Null rate	0.220	0.509
Blank rate	0.018	0.340
Valid rate	0.038	0.485

Interpretation

Blank-rate effect is not robust to the adversarial-only specification. It may reflect a behavioral response to the overall *breadth* of electoral litigation rather than adversarial challenges per se. Further investigation warranted.

Voter Behavior — Forest Plot (All Specs)



Thick bars: 90% CI. Thin bars: 95% CI. SE clustered by principal electoral zone.

Outcome: Δ (vote share accruing to the 2020 incumbent candidate), 2024 – 2020

Specification	Coef.	SE	<i>t</i>	<i>p</i>	<i>N</i>
Baseline (state FE)	−0.065	0.065	−1.00	0.316	5,560
Baseline, single-zone	−0.065	0.067	−0.97	0.331	5,371
Full controls	−0.056	0.064	−0.87	0.383	5,560
≤200k voters	−0.065	0.066	−0.99	0.323	5,338
>200k voters	−0.063	0.249	−0.25	0.801	222

Adversarial-only (narrow) specification. SE clustered by principal electoral zone.

- Negative across all specifications: judicialization marginally reduces the incumbent's vote share
- Estimates are modest (−6 to −7 pp) and imprecise ($p > 0.31$)
- Consistent with diffuse rather than targeted litigation patterns

Winner Identity: New vs. Returning Candidate

Outcome: $\Delta(\text{winner is a candidate who did not run in 2020}), 2024 - 2020$

Specification	Coef.	SE	<i>t</i>	<i>p</i>	<i>N</i>
Baseline (state FE)	+0.182	0.104	+1.75	0.082	5,560
Baseline, single-zone	+0.196	0.108	+1.82	0.070	5,371
Full controls	+0.173	0.103	+1.69	0.092	5,560
$\leq 200\text{k}$ voters	+0.185	0.106	+1.75	0.081	5,338
$> 200\text{k}$ voters	+0.298	0.529	+0.56	0.574	222

Adversarial-only specification. SE clustered by principal electoral zone.

- Positive and marginally significant in adversarial-only spec ($p \approx 0.08$): more adversarial litigation associated with a higher probability that the winner is a new entrant
- Effect is not robust in the broad specification ($p > 0.98$) — suggests DRAP and RRC dilute this signal
- Direction is consistent across specifications; warrants further investigation

Pattern across incumbency channels:

Judicialization $\uparrow$

$\Rightarrow$ Incumbent vote share:
-6.5 pp ($p = 0.32$)

$\Rightarrow$ Winner is new entrant:
+18 pp ($p = 0.08^*$)

Suggestive: adversarial challenges may favour new entrants

Key implication: if the winner-is-new result is real, adversarial litigation waves may not simply reinforce incumbents — they may open space for challengers who are not yet embedded in litigation targets. Requires replication and candidate-level analysis.

Not robust across variants

Winner-is-new result ($p = 0.08$) appears only in adversarial-only spec; disappears with broad instrument ($p > 0.98$). Treat as exploratory.

Systemic, Not Targeted

Incumbent vote share is negative but imprecise across all specs. Adversarial litigation waves diffuse across the competitive field rather than targeting specific incumbents — consistent with the network-diffusion instrument logic.

Hypothesis: lawsuits may matter more in close races, where a small distraction can tip the balance.

Subgroup split: 2020 margin above/below median (≈ 13.4 pp)

- **High competition** (close in 2020): more room for lawsuits to matter
- **Low competition** (landslide in 2020): effects may be muted

Subsample IV

Subsample	Coef.	SE	<i>p</i>
Close races ($\leq$ median)	+0.022	0.041	0.584
Landslide ($>$ median)	+0.085	0.059	0.151

Outcome: Δ electoral margin. Broad specification. Adversarial-only produces similar pattern.

Hypothesis: if lawsuits disproportionately target incumbents (more visible, longer record to challenge), the effect should be larger in municipalities where the 2020 winner re-runs.

Subgroup split: whether the 2020 mayor appears on the 2024 ballot

- **Incumbent runs** (54% of municipalities): mixed race
- **Open seat** (46%): no incumbent; different dynamics

Caveat: incumbency in 2024 is itself a potential outcome; this is a descriptive heterogeneity exercise, not a clean causal subgroup.

Subsample IV

Subsample	Coef.	SE	<i>p</i>	<i>N</i>
Incumbent runs	+0.060	0.054	0.268	2,894
Open seat	+0.030	0.041	0.461	2,466

Outcome: Δ electoral margin. Broad specification.

Extended outcomes

Outcome	Coef.	SE	<i>t</i>	<i>p</i>	<i>N</i>
Δ Winner ideology (2024 – 2020)	–0.168	0.205	–0.82	0.412	5,387
Pedersen electoral volatility	–0.010	0.038	–0.25	0.801	5,491

Broad specification, baseline state FE. SE clustered by principal electoral zone.

- **Winner ideology:** no systematic left-right shift — lawsuits do not reshape the ideological composition of winners ($p = 0.41$)
- **Pedersen volatility:** overall party-level vote-share volatility unaffected ($p = 0.80$) — judicialization does not generate broader electoral churning

Does judicialization shift the share of votes going to female candidates?

Specification	<i>Broad (F = 19.6)</i>			<i>Narrow / preferred (F = 10.4)</i>		
	Coef.	SE	p	Coef.	SE	p
Baseline (state FE)	−0.093	0.048	0.052	−0.067	0.052	0.198
Baseline, single-zone	−0.089	0.049	0.073	−0.059	0.054	0.274
Full controls	−0.101	0.044	0.023	−0.084	0.047	0.072
≤200k voters	−0.089	0.049	0.068	−0.059	0.053	0.265
>200k voters	−0.082	0.165	0.619	−0.076	0.188	0.686

Outcome: Δ female vote share 2024–2020. SE clustered by electoral zone.

Pattern

Consistently negative across both variants. More pronounced under the broad spec ($p = 0.052$ in baseline); narrower instrument attenuates toward insignificance ($p = 0.20$). Full-controls estimate reaches $p = 0.023$.

Interpretation caveat

Full-controls significance may reflect post-treatment bias: 2020 candidate composition controls could absorb baseline gender endowment. Direction is suggestive but not robust.

Does judicialization change the probability that a woman wins the mayoral race?

Specification	Broad ($F = 19.6$)			Narrow / preferred ($F = 10.4$)		
	Coef.	SE	p	Coef.	SE	p
Baseline (state FE)	-0.123	0.063	0.052	-0.047	0.066	0.474
Baseline, single-zone	-0.125	0.066	0.057	-0.049	0.069	0.479
Full controls	-0.126	0.060	0.037	-0.063	0.063	0.318
$\leq 200k$ voters	-0.116	0.064	0.068	-0.041	0.067	0.544
$> 200k$ voters	-0.087	0.254	0.731	-0.126	0.285	0.658

Outcome: Δ winner is female 2024–2020 (binary). SE clustered by electoral zone.

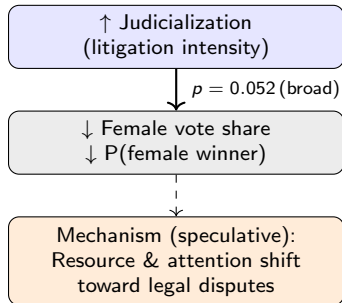
Pattern

Negative in broad spec ($p = 0.052$); disappears in adversarial-only ($p = 0.47$). Direction consistent with female vote share finding. The broad-spec estimate may partly reflect DRAP/RRC variation.

Comparison

Direction mirrors female vote share result. Neither reaches $p < 0.05$ under the preferred adversarial-only specification. Gender effects are not robustly identified.

Interpretation: Gender Results



Overall verdict

No robust evidence of a gender effect. Point estimates are consistently negative but do not survive the switch to the adversarial-only specification. Results sensitive to the administrative filings included in the instrument.

Directional story

If real: adversarial litigation waves *may* disadvantage female candidates — through resource diversion, hostile-environment deterrence, or selection toward incumbents (predominantly male). Requires a focused study with candidate-level data.

Questions:

- Does judicialization deter candidate entry overall?
- Does it select for career politicians better equipped to handle legal challenges?
- Does it shift vote share toward candidates with more prior experience?

IV Results (adversarial-only)

Outcome	Coef.	SE	<i>p</i>
Δ Log n candidates	-0.052	0.047	0.265
Δ New cand. vote share	+0.132	0.077	0.085
Δ Incumb. vote share	-0.065	0.065	0.316

*Broad spec: log candidates -0.066 ($p=0.092$),
new cand. vote -0.010 ($p=0.848$).*

Art. 29-II CF/88: municipalities with $> 200,000$ registered voters hold a mandatory second round if no candidate wins $> 50\%$

Two subsamples:

- $\leq 200k$ voters: 5,338 municipalities — strictly single-round regime, plurality wins
- $> 200k$ voters: 222 municipalities — conditional second round; larger, more urban, more media attention

Why this matters: homogeneity within each regime; the $\leq 200k$ group faces the same constitutional incentive structure throughout

First stage by subsample

Subsample	Adv. F	Broad F	N
$\leq 200k$	10.2	19.3	5,338
$> 200k$	0.7	1.1	222

Large-city subsample ($> 200k$): instrument loses power in both variants; IV estimates unreliable. Instrument identification is concentrated among smaller municipalities.

7 → 14 variable expansion. Extended controls add:

- ENP 2016 (pre-trend competitiveness)
- Candidate composition: female share, nonwhite share, higher-education share (2020)
- Party structure: log(party count), log(coalition count) (2020)
- Voter behavior: turnout, null share (2020)
- Candidate experience: share first-time, share career politicians (2020)

Note: some of these are outcome-adjacent (2020 candidate composition may reflect prior judicialization). Treat as sensitivity only.

7 vs. 14 Controls (Electoral Margin)

Controls	Coef.	SE	<i>p</i>	F
<i>Adversarial-only</i>				
7 baseline	+0.057	0.051	0.268	10.4
14 extended	+0.052	0.049	0.291	10.7
<i>Broad</i>				
7 baseline	+0.062	0.043	0.151	19.6
14 extended	+0.059	0.042	0.162	19.7

Standard sample ($N = 5,560$): includes municipalities spanning multiple zones; clustering approximated by principal zone

Single-zone sample ($N = 5,371$, 96.6%): municipality = zone = cluster; correspondence is exact

Why it matters: multi-zone municipalities tend to be larger and may have different legal and political dynamics; exact clustering removes the approximation error in inference

All vs. Single-Zone (Adversarial-only)

	All	Single-zone
Δ Runner-up	-0.019	-0.017
Δ Margin	+0.057	+0.056
Δ Winner	+0.038	+0.040
Δ Majority	+0.113	+0.115
F-stat	10.4	10.3
N	5,560	5,371

Estimates essentially identical. Robust to sample composition.

1 **Credible identification:**

Bartik IV valid across two topic-scope variants. Adversarial-only ($F \approx 10$, preferred) and broad ($F \approx 20$). First-stage stable across specifications.

2 **Winner consolidation (directional):**

Both variants point toward more concentrated outcomes: higher winner majority probability, wider margin, lower candidate count. Direction consistent; statistically imprecise at $\alpha = 0.05$ with the adversarial-only specification.

3 **Blank vote rate (broad spec only):**

Significant positive effect (+1.3 pp, $p = 0.018$) under the broader instrument. Not robust in adversarial-only. Suggests voters respond to judicial breadth with expressive abstention.

Main Takeaway

We provide credibly identified estimates of the electoral effects of judicialization using a Bartik shift-share instrument.

The winner consolidation pattern is directionally consistent and robust to instrument scope. Statistical significance depends on whether mandatory administrative filings are included in the topic mix. This sensitivity is itself informative: administrative litigation breadth — not just adversarial challenges — appears to matter for voter behavior.

Contributions:

- First Bartik IV applied to electoral judicialization; establishes a blueprint for identifying causal electoral effects of litigation
- Large-scale evidence from 5,560 Brazilian municipalities across two election cycles
- Systematic distinction between administrative and adversarial litigation — a methodological contribution to measurement in the electoral law literature
- Two-variant robustness design reveals that administrative filing breadth and adversarial challenges operate through distinct channels

Next Steps

- AKM (2019, QJE) standard error correction for few-effective-shocks designs
- Romano–Wolf multiple testing correction across outcome families
- Post-election timing placebo (cases filed after election day)
- Heterogeneity by TRE judge composition
- Case *outcomes* (ruling for vs. against) as a second instrument
- Longer panel: 2012–2024 for trend identification

Appendix A: Full First-Stage Table — Both Variants I

Panel A: Broad instrument (all competition topics)

Specification	Coef.	SE	<i>t</i>	F-stat	<i>N</i>
Baseline (state FE)	1.792	0.405	4.42	19.6	5,560
Baseline, single-zone	1.801	0.422	4.27	18.2	5,371
Full controls	1.799	0.406	4.44	19.7	5,560
≤200k voters	1.869	0.426	4.39	19.3	5,338
>200k voters	0.799	0.779	1.03	1.1	222

Panel B: Adversarial-only instrument (preferred, excl. mandatory filings)

Specification	Coef.	SE	<i>t</i>	F-stat	<i>N</i>
Baseline (state FE)	0.481	0.149	3.23	10.4	5,560
Baseline, single-zone	0.497	0.155	3.21	10.3	5,371
Full controls	0.491	0.151	3.26	10.7	5,560
≤200k voters	0.493	0.155	3.19	10.2	5,338
>200k voters	0.485	0.566	0.86	0.7	222

SE clustered by principal electoral zone. $F = (t\text{-stat on excluded instrument})^2$.

Appendix B: Full IV Table — Competition and Voter Behavior I

Adversarial-only specification (preferred). Baseline, N = 5,560, clusters = 2,187.

Outcome	Coef.	SE	<i>t</i>	<i>p</i>
<i>Competition</i>				
Δ Runner-up	−0.019	0.026	−0.75	0.456
Δ Margin	+0.057	0.051	+1.11	0.268
Δ Winner vote	+0.038	0.030	+1.26	0.207
Δ Majority	+0.113	0.100	+1.13	0.259
Δ Log candidates	−0.052	0.047	−1.12	0.265
<i>Voter behavior</i>				
Δ Turnout rate	−0.008	0.007	−1.08	0.281
Δ Null rate	−0.003	0.005	−0.66	0.509
Δ Blank rate	+0.004	0.004	+0.95	0.340
Δ Valid vote rate	−0.008	0.012	−0.70	0.485
<i>Broad specification (all topics). Baseline, same sample.</i>				
Δ Runner-up	−0.031	0.022	−1.42	0.157
Δ Margin	+0.062	0.043	+1.44	0.151
Δ Winner vote	+0.031	0.025	+1.24	0.213
Δ Majority	+0.140	0.090	+1.55	0.122
Δ Blank rate	+0.013	0.005	+2.37	0.018

Both specifications, baseline state FE, N = 5,560.

Outcome	<i>Broad</i>			<i>Adversarial-only</i>		
	Coef.	SE	<i>p</i>	Coef.	SE	<i>p</i>
Δ Female vote share	−0.093	0.048	0.052	−0.067	0.052	0.198
Δ Nonwhite vote share	−0.039	0.042	0.348	−0.048	0.067	0.480
Δ New cand. vote share	−0.010	0.050	0.848	+0.132	0.077	0.085
Δ Incumb. vote share	+0.010	0.049	0.844	−0.065	0.065	0.316
Δ Winner is female	−0.123	0.063	0.052	−0.047	0.066	0.474
Δ Winner is new	−0.002	0.073	0.982	+0.182	0.104	0.082
Δ Winner ideology	−0.168	0.205	0.412			
Pedersen volatility	−0.010	0.038	0.801			
SE clustered by principal electoral zone.						

Competition-relevant adversarial case criteria:

- 1 Mayoral candidate (*prefeito*) is the respondent (passive party)
- 2 Filed before election day (filing date < election date)
- 3 Subject is not a mandatory administrative filing (excl. RRC and DRAP in preferred spec)

Examples of adversarial case subjects:

Abuso do Poder Econômico, Abuso do Poder Político, Captação Ilícita de Sufrágio, Propaganda Eleitoral em Lugar Proibido, Propaganda Irregular, Conduta Vedada a Agente Público, Uso Indevido dos Meios de Comunicação, Impugnação de Registro de Candidatura

Case Volume (2020 / 2024)

Category	2020	2024
Mandatory registration (RRC)	~994k	~841k
Coalition registration (DRAP)	~137k	~118k
Adversarial competition cases	~385k	~340k
<i>TSE administrative records. Pre-election, first-instance cases.</i>		

Figure E1: Distribution of Bartik IV

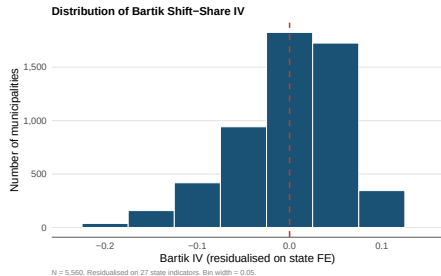
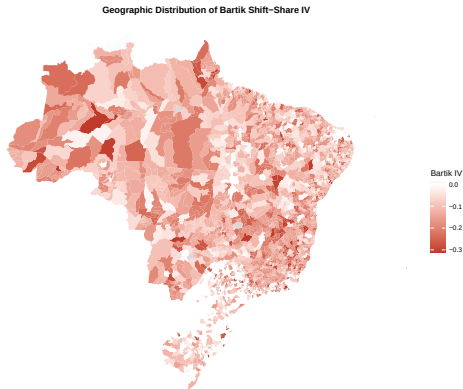


Figure E2: Geographic distribution



Shading = raw Bartik IV value (winsorised at 1st/99th percentile). Red = lower-than-average IV (topic mix predicts decline). Blue = higher-than-average IV. Grey = municipalities not in an.

Adversarial-only (preferred) instrument. IV residualised on state FE for histogram. Map: raw Bartik IV by municipality (grey = not in sample).

Appendix F: Balance Table — Instrument on Pre-Determined Characteristics I

Adversarial-only instrument regressed on each covariate, state FE, SE clustered by zone.

Covariate	Coef.	SE	<i>p</i>
Log population (2010)	−0.543	0.256	0.034
Urban share (2010)	−0.024	0.028	0.385
Log income per capita (2010)	+0.164	0.049	0.001
Margin 2016	+0.089	0.041	0.030
ENP 2016	−0.256	0.142	0.072
Turnout rate 2020	+0.040	0.012	0.001
Female candidate share 2020	−0.075	0.046	0.101
Nonwhite candidate share 2020	−0.050	0.067	0.458
Log n candidates 2020	−0.127	0.084	0.129

3 of 9 covariates: $p < 0.10$. Correlation with log pop, income, turnout reflects that larger, richer, higher-turnout municipalities had more diverse litigation.

All pre-determined controls included in main regressions.

All main results estimated in **R** using
`fixest::feols()` v0.14.1:

IV formula (adversarial-only):

```
feols(outcome ~ controls | state |  
      delta_log1p_adversarial ~  
      bartik_iv_adversarial,  
      data, cluster = ~zone_id)
```

Benchmark: Ash et al. (2025) Stata:
`ivreghdfe y (endog = instr),
absorb(state) cl(zone)`

Replication summary

Outcome	Coef.	SE	<i>p</i>
<i>Adv.-only: $F = 10.4$, $N = 5,560$</i>			
Δ Majority	+0.113	0.100	0.259
Δ Blank rate	+0.004	0.004	0.340
<i>Broad: $F = 19.6$, $N = 5,560$</i>			
Δ Majority	+0.140	0.090	0.122
Δ Blank rate	+0.013	0.005	0.018

SE clustered by principal electoral zone.

Art. 29-II, Constituição Federal 1988:

“eleição do Prefeito e do Vice-Prefeito realizada no dia 1.^o de outubro do ano anterior ao do término do mandato dos que devam suceder, aplicadas as regras do art. 77, no caso de Municípios com mais de duzentos mil eleitores.”

Implication: the 200,000 registered voter threshold

determines whether a municipality follows plurality rule (one round) or the two-round runoff rule. This creates a sharp institutional discontinuity that motivates our population subsample as a homogeneity check.

Sample Split (2024 Registered Voters)

	<i>N</i>	Share
$\leq 200,000$	5,338	96.0%
$> 200,000$	222	4.0%
Total	5,560	100%